

1. Name of the medicinal product

DIACOL, 27 mg/15 ml, syrup

2. Qualitative and quantitative composition

27 mg dextromethorphan hydrobromide per 15 ml of syrup.

For excipients, see 6.1.

3. Pharmaceutical form

Syrup.

4. Clinical particulars**4.1. Therapeutic indications**

Symptomatic treatment of non-productive cough due to irritation of upper and lower respiratory tract.

4.2. Posology and administration

Unless your doctor advises otherwise, DIACOL should be taken as follows:

Adults and adolescents: 5 to 10 ml of dextromethorphan 4 times a day. Maximum daily dose is 60 ml.

Children 2 to 12 years: dosage according to medical prescription. Children aged 2 to 6 years old, maximum daily dose is 15 ml. Children aged 6 to 12 years old, maximum daily dose is 30 ml;

Children under 2 years: Not recommended for this age group.

DIACOL may be administered every 4 hours, reducing the amount of syrup per intake, not exceeding the maximum daily dose.

Do not exceed the recommended doses or use for long periods.

Take it alone or diluted in a small amount of water.

4.3. Contraindications

Hypersensitivity to any of the ingredients. Productive cough. Concomitant treatment with monoamine oxidase inhibitors (MAO).

4.4. Special warnings and special precautions for use

DIACOL should be used with caution in asthmatic or atopic patients, since it may promote histamine release. And also in sedated or debilitated patients.

In patients with hepatic impairment it may occur metabolism changes.

Cases of dextromethorphan abuse and dependence have been reported. Caution is particularly recommended for adolescents and young adults as well as in patients with a history of drug abuse or psychoactive substances.

If cough persists for longer than 1 week, tends to recur, or is accompanied by high fever, skin eruption or strong headache, a physician should be consulted.

DIACOL contains 121.64 g sucrose per 200 ml of syrup. A 15 ml-dose contains approximately 9.1 g sucrose. Do not use in case of inherited fructose intolerance, glucose-galactose malabsorption syndrome or sucrase / isomaltase deficiency.

DIACOL contains 2.75 g of ethanol 96% per 200 ml of syrup. A 15 ml-dose contains approximately 0.20 g of ethanol 96%. It can be harmful for those suffering from liver disease, alcoholism, epilepsy, brain injury or disease, as well as for pregnant women and children. May modify or increase the effect of other medicines.

Dextromethorphan is metabolised by hepatic cytochrome P450 2D6. The activity of this enzyme is genetically determined. About 10% of the general population are poor metabolisers of CYP2D6. Poor metabolisers and patients with concomitant use of CYP2D6 inhibitors may experience exaggerated and/or prolonged effects of dextromethorphan. Caution should therefore be exercised in patients who are slow metabolizers of CYP2D6 or use CYP2D6 inhibitors (see also section 4.5).

Serotonin Syndrome

Serotonergic effects, including the development of a potentially life-threatening serotonin syndrome, have been reported for dextromethorphan with concomitant administration of serotonergic agents, such as selective serotonin re-uptake inhibitors (SSRIs), drugs which impair metabolism of serotonin (including monoamine oxidase inhibitors (MAOIs)) and CYP2D6 inhibitors.

Serotonin syndrome may include mental-status changes, autonomic instability, neuromuscular abnormalities, and/or gastrointestinal symptoms.

If serotonin syndrome is suspected, treatment with DIACOL should be discontinued.

Paediatric population

Serious adverse events may occur in children in case of overdose including neurological disorders. Caregivers should be advised not to exceed the recommended dose.

4.5. Interactions with other medicinal products and other forms of interaction

Monoamine (MAO) oxidase inhibitors:

Concurrent use of DIACOL with MAO inhibitors may promote adrenergic crisis, coma, vertigo, excitability, hypertension, hyperpyrexia, intracerebral haemorrhage, lethargy, nausea, psychotic behaviours, muscle spasms, tremors and, eventually death and therefore concomitant administration is not recommended.

An interval of 14 days should be respected between withdrawal of monoamine oxidase inhibitors and DIACOL intake.

Medicines acting upon the central nervous system:

Effects of medicines acting upon the central nervous system may be enhanced when used concurrently with dextromethorphan.

CYP2D6 inhibitors:

Dextromethorphan is metabolized by CYP2D6 and has an extensive first-pass metabolism. Concomitant use of potent CYP2D6 enzyme inhibitors can increase the dextromethorphan concentrations in the body to levels multifold higher than normal. This increases the patient's risk for toxic effects of dextromethorphan (agitation, confusion, tremor, insomnia, diarrhoea and respiratory depression) and development of serotonin syndrome. Potent CYP2D6 enzyme inhibitors include fluoxetine, paroxetine, quinidine and terbinafine. In concomitant use with quinidine, plasma concentrations of dextromethorphan have increased up to 20-fold, which has increased the CNS adverse effects of the agent. Amiodarone, flecainide and propafenone, sertraline, bupropion, methadone, cinacalcet, haloperidol, perphenazine and thioridazine also have similar effects on the metabolism of dextromethorphan. If concomitant use of CYP2D6 inhibitors and dextromethorphan is necessary, the patient should be monitored and the dextromethorphan dose may need to be reduced.

4.6. Pregnancy and lactation

Although there are no reports of foetal problems secondary to the use of DIACOL during pregnancy or interference on the reproduction system DIACOL is not recommended during pregnancy. Also in lactation.

4.7. Effects on ability to drive and use machines

Diacol has no influence on the ability to drive and use machines.

4.8. Undesirable effects

Usually, side effects due to DIACOL are rare, and may appear as slight gastrointestinal disturbances, namely nausea and vomiting, stomach pains, drowsiness and vertigo.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions (see details below).

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4.9. Overdose

Symptoms and signs:

Dextromethorphan overdose may be associated with nausea, vomiting, dystonia, agitation, confusion, somnolence, stupor, nystagmus, cardiotoxicity (tachycardia, abnormal ECG including QTc prolongation), ataxia, urinary retention, toxic psychosis with visual and auditory hallucinations, hyperexcitability.

In the event of massive overdose the following symptoms may be observed: coma, respiratory depression, convulsions.

Management:

-Activated charcoal can be administered to asymptomatic patients who have ingested overdoses of dextromethorphan within the preceding hour.

-For patients who have ingested dextromethorphan and are sedated or comatose, naloxone, in the usual doses for treatment of opioid overdose, can be considered. Benzodiazepines for seizures and benzodiazepines and external cooling measures for hyperthermia from serotonin syndrome can be used.

5. Pharmacological properties

5.1. Pharmacodynamic properties

ATC Classification: R05D A09.

Dextromethorphan is a central acting antitussive. It is an isomer of levorphanol, a codeine analogue. Unlike the other isomers, dextromethorphan has no analgesic or addictive properties, and does not act by means of opiate receptors. Its antitussive properties result from its direct action on the cough centre in the medulla, raising the threshold for the cough reflex. Its antitussive strength is similar to codeine, persisting its effects for 6 hours. Contrarily to codeine, its sedative action is practically negligible, since it only occurs at extremely high doses.

In therapeutic dosage, it does not inhibit the ciliary function of airways, thus providing a clinically advantage over many other cough suppressants.

5.2. Pharmacokinetic properties

Dextromethorphan is rapidly absorbed from the gastrointestinal tract. T_{max} is 2 to 2.5 hours and C_{max} after an oral dose of 60 mg is 5.2 ng/ml. It exerts its antitussive effect in 15-30 minutes after oral administration and the duration of action is approximately 5-6 hours.

Dextromethorphan is distributed to CNS, reaching detectable concentrations on cerebrospinal fluid/plasma concentrations ranging between 32.8 and 80%.

Dextromethorphan undergoes rapid and extensive first-pass metabolism in the liver after oral administration. Genetically controlled O-demethylation (CYD2D6) is the main determinant of dextromethorphan pharmacokinetics in human volunteers.

It appears that there are distinct phenotypes for this oxidation process resulting in highly variable pharmacokinetics between subjects. Unmetabolised dextromethorphan, together with the three demethylated morphinan metabolites dextrorphan (also known as 3-hydroxy-N-

methylmorphinan), 3- hydroxymorphinan and 3-methoxymorphinan have been identified as conjugated products in the urine.

Dextrophan, which also has antitussive action, is the main metabolite. In some individuals metabolism proceeds more slowly and unchanged dextromethorphan predominates in the blood and urine.

Dextromethorphan and its demethylated metabolites are excreted primarily in urine. Elimination half-life of dextromethorphan is 1.4 to 3.9 hours and of dextrophan of 3.4 to 5.6 hours.

5.3. Preclinical safety data

Dextromethorphan is a currently used medicine, included in practically all pharmacopoeias.

6. Pharmaceutical particulars

6.1. List of excipients

Sucrose, anhydrous citric acid, methyl-paraben; liquid orange flavour; ethanol 96%, purified water.

6.2. Incompatibilities

In absence of incompatibilities this medicinal product must not be mixed with other medicinal products.

6.3. Shelf-life

5 years.

After first opening: 4 years.

Do not use this medicine after the expiry date, which is stated on the bottle and the carton after EXP. The expiry date refers to the last day of that month.

6.4. Special precautions for storage

No special precautions for storage are needed.

6.5. Nature and contents of container

Carton packs containing a glass bottle, type III, amber colour, and a leaflet. The bottle contains 200 ml-syrup. Each box contains a dosing cup.

6.6. Instructions for use and handling

No special requirements.

7. Marketing Authorisation Holder

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Portugal

8. Marketing Authorisation Number(s)

19560

9. Date of first Authorisation

30th April 2004

10. Date of revision of the text